

VeriFund AI - Product Requirements Document (PRD)

High-Precision Financial Advisor Copilot & Compliance Verification Platform

1. Executive Summary

VeriFund AI is an enterprise-grade, retrieval-augmented generation (RAG) copilot designed for financial advisors, wealth managers, and compliance officers.

Financial advisory firms maintain tens of thousands of pages of rapidly changing market research, fund prospectuses, factsheets, and regulatory disclosures. Today, advisors navigate disconnected document repositories and static PDF portals while interacting with clients.

Because manual verification is slow and standard generic AI models hallucinate facts, advisors risk delivering outdated or non-compliant financial guidance.

VeriFund AI solves this problem by combining:

- **Tree-Structured & Hierarchical Document Ingestion:** Preserving parent-child regulatory context and complex 2D financial tables.
- **Temporal & Status-Aware Filtering:** Ensuring answers strictly originate from active, compliance-approved document versions.
- **Split-Screen Interactive Grounding:** Linking every generated answer to page-level, highlighted bounding boxes in original PDF source documents.
- **Automated Guardrails & Disclaimers:** Running real-time faithfulness evaluations and auto-injecting mandatory regulatory disclaimers.

2. Problem Statement

Wealth management and financial advisory firms operate in heavily regulated environments. Firms face severe risks due to information latency and document silos.

Advisor Challenges

- **Information Fragmentation:** Critical fund data is scattered across quarterly factsheets,

prospectuses, market updates, and compliance bulletins.

- **Verification Friction:** Finding exact clauses regarding liquidity limits, exit fees, or accreditation criteria during client meetings takes minutes of manual searching.
- **Client Translation Burden:** Converting dense legal prospectus clauses into accessible client summaries while maintaining compliance is difficult and error-prone.

Compliance & Legal Challenges

- **Regulatory Penalties:** Giving non-compliant or outdated advice exposes firms to severe fines, client litigation, and loss of advisory licenses.
- **Version Drift:** Advisors frequently rely on saved local copies or superseded factsheets that contain outdated expense ratios and historical returns.
- **Absence of Auditability:** Standard conversational tools do not provide an immutable, verifiable link between the generated advice and the exact document version used at that point in time.

Technical & AI Challenges

- **Hallucination Risk:** Generic LLMs invent numbers, drop negative constraints, or blend numbers from different fiscal quarters.
- **Tabular & Layout Parsing Failures:** Traditional naive chunking breaks multi-column tables, footnotes, and fee schedules in financial PDFs.

3. Product Vision

"The Bloomberg Terminal + Compliance Engine for Advisory Communication"

VeriFund AI aims to answer any complex fund or regulatory question within seconds, backed by verifiable visual citations and guaranteed zero-hallucination compliance.

4. Goals and Objectives

1. **Zero-Hallucination Retrieval:** Guarantee that 100% of facts, metrics, and policy rules in responses are grounded in verified firm documents.
2. **Instant Visual Verification:** Enable advisors to inspect and verify the exact source PDF page and bounding box in under 1 second via a split-screen interface.
3. **Temporal Compliance Integrity:** Enforce active document version control with "As-Of" historical auditing capabilities.

4. **Workflow Acceleration:** Reduce time spent verifying fund metrics and preparing compliant client communication by over 70%.

5. Scope & Feature Boundaries

Included in MVP Scope

- **Secure Document Ingestion Pipeline:** Upload and structural parsing of financial PDFs.
- **Tree-Structured & Tabular Indexing:** Hierarchical chunking and dual-representation table extraction.
- **Hybrid Search Engine:** Dense vector similarity + Sparse BM25 keyword matching.
- **Split-Screen Verification Interface:** Interactive chat alongside a synchronized PDF reader.
- **Temporal Version & Status Management:** Metadata filtering for document status and effective dates.
- **Automated Faithfulness Scoring:** Real-time evaluator agent checking generated text against retrieved source chunks.
- **Mandatory Disclosure Engine:** Automatic injection of SEC/FINRA-compliant disclaimers.
- **Client-Ready Translation Toggle:** Switching between detailed "Advisor Mode" and simplified "Client Summary Mode."

Excluded from MVP Scope (Out of Scope)

- Direct Brokerage/Custodian API Integrations (e.g., Plaid, Schwab).
- Automated Voice Streaming / Call Interception (deferred to Post-MVP).
- Autonomous Outbound Email Sending (MVP outputs formatted copyable text only).
- Multi-Tenant Enterprise SSO (MVP uses standard JWT authentication).
- Smart Contract / Blockchain Audit Ledger (deferred to enterprise roadmap).

6. Target Users & User Personas

1. Financial Advisor / Planner	2. Compliance & Risk Officer
Needs: - Instant retrieval during calls	Needs: - Ensure advisors use approved materials

1. Financial Advisor / Planner	2. Compliance & Risk Officer
- Accurate fee & return numbers - Client-ready compliant text	- Audit trail of advisor queries - Fast deprecation of outdated documents

7. Product Modules & Functional Specifications

Module 1: Document Ingestion & Structural Tree Parser

- **Document Upload:** Support for multi-page financial PDFs.
- **Layout-Aware Extraction:** Detects headings, sections, footnotes, and page numbers. Preserves full hierarchical path in chunk metadata.
- **Dual-Representation Tabular Processing:** Extracts tables into clean JSON/Markdown matrices. Generates semantic summary for embedding while storing raw table for calculation.
- **Contextual Enrichment:** Automatically prepends document title, effective date, and fund ticker to leaf chunks.

Module 2: Hybrid Search & Contextual Retrieval Engine

- **Hybrid Retrieval:** Executes parallel vector search and BM25 keyword matching.
- **Hierarchical Parent Expansion:** Automatically extracts enclosing parent sections when an atomic leaf chunk matches.
- **Re-Ranking Pipeline:** Cross-encoder re-ranker to score and filter chunks.

Module 3: Advisor Copilot & Split-Screen Grounding UI

- **Split-Screen Workspace:** Left Pane for Chat/Prompt, Right Pane for high-speed PDF Viewer.
- **Interactive Deep Citations:** Clicking citation pills automatically scrolls to the page and draws a highlighted bounding box over the source paragraph or table.
- **Tone & Persona Toggle:** Toggle between Advisor Mode and Client-Ready Mode.

Module 4: Faithfulness Critic & Automated Disclaimer Engine

- **Faithfulness Evaluation Pass:** Computes a Groundedness Score (0–100%) against retrieved chunks.
- **Safety Guardrail:** Rejects responses falling below 85% confidence score.

- **Regulatory Disclaimer Engine:** Appends mandatory disclosure paragraphs based on asset categories.

Module 5: Document Versioning & Compliance Audit Manager

- **Metadata Lifecycle:** Documents tagged as APPROVED, DRAFT, SUPERSEDED, or EXPIRED.
- **As-Of Date Selector:** Query knowledge base as it existed on any past date.
- **Audit Logging:** Logs every query, retrieved chunk ID, model output, and user feedback.

8. Database Design

Schema Field Specifications:

documents Table

- id: UUID (Primary Key)
- organization_id: UUID (Foreign Key)
- title: VARCHAR(255)
- file_url: TEXT (S3 / Cloud Storage URL)
- status: ENUM (APPROVED, DRAFT, SUPERSEDED, EXPIRED)
- effective_date: DATE
- expiration_date: DATE (Nullable)
- version: VARCHAR(50)
- asset_class: VARCHAR(100)
- created_at: TIMESTAMP

document_chunks Table

- id: UUID (Primary Key)
- document_id: UUID (Foreign Key)
- parent_chunk_id: UUID (Nullable)
- chunk_type: ENUM (HEADER, SECTION, LEAF_PARAGRAPH, TABLE_SUMMARY, RAW_TABLE)
- content: TEXT
- page_number: INTEGER
- bounding_box_json: JSONB ({x1, y1, x2, y2})
- tree_path: VARCHAR(500)
- embedding_id: VARCHAR(255)

query_logs Table

- id: UUID (Primary Key)
- user_id: UUID (Foreign Key)

- query_text: TEXT
- response_text: TEXT
- retrieved_chunk_ids: JSONB
- faithfulness_score: FLOAT
- as_of_date: DATE
- disclosures_applied: JSONB
- created_at: TIMESTAMP

9. API Design

- POST /api/v1/auth/login - Authenticate user and return JWT.
- GET /api/v1/auth/me - Retrieve current user profile.
- POST /api/v1/documents/upload - Upload PDF and trigger ingestion.
- GET /api/v1/documents - List documents with filters.
- PATCH /api/v1/documents/:id/status - Update document lifecycle status.
- POST /api/v1/copilot/query - Primary query endpoint supporting streaming response.
- POST /api/v1/copilot/evaluate - Internal endpoint to run faithfulness critic.

10. System Architecture

- **Frontend User Interface:** React + TypeScript Web App with split-screen viewer and highlight renderer.
- **Backend API Gateway:** FastAPI handles auth, rate-limiting, and routes requests to the Query Orchestrator.
- **RAG Pipeline Core:** Executes Hybrid Search, Tree Context Expansion, and triggers the Faithfulness Critic Agent.
- **Data Storage:** PostgreSQL for relational data and logs, Vector DB (Chroma/Qdrant) for embeddings, and Cloud Storage for original PDFs.
- **LLM Provider:** External API call (e.g., Gemini API) for text synthesis and validation.

11. Technology Stack

Layer	Selected Technology
Frontend	React 19, TypeScript, Vite, Tailwind CSS, shadcn/ui, react-pdf-highlighter, Lucide React
Backend	Python 3.11+, FastAPI, Pydantic v2, LlamaIndex/LangChain, Rank-BM25
Ingestion & OCR	LlamaParse, Unstructured.io, pdfplumber
Database & Vector	PostgreSQL, ChromaDB/Qdrant, Redis
AI Models	Gemini 1.5 Pro/Flash, text-embedding-3-small
Deployment	Vercel (Frontend), Render (Backend)

12. Security & Compliance Requirements

- **Role-Based Access Control (RBAC):** Advisor, Compliance Officer, and Admin tiers.
- **Data Protection & Encryption:** TLS 1.3 enforced, AES-256 encryption at rest.
- **Strict Source Isolation:** Prompt engineering prevents reliance on pre-trained numerical knowledge.
- **Sanitization & Input Validation:** XSS filtering, parameterized SQL, and rate limiting.

13. Non-Functional Requirements (NFRs)

- **Performance & Latency:** Less than 1.2 seconds to first token. Full query turnaround under 3.0 seconds. PDF coordinate jump under 250 milliseconds.
- **Reliability & Availability:** 99.9% uptime target. Graceful fallback on ungrounded queries.
- **Scalability:** Parallel async ingestion of 100+ page prospectuses without blocking chat requests.

14. Development Roadmap

Phase 1: Foundation (Week 1)	Phase 2: Ingestion (Week 2)	Phase 3: UI & Viewer (Week 3)	Phase 4: Compliance (Week 4)
- FastAPI setup - PostgreSQL auth - ChromaDB init - Schema creation	- PDF parser setup - Table extraction - Hierarchical RAG - BM25 integration	- Split-screen layout - PDF viewer + highlight - Chat UI streaming	- Faithfulness critic - Auto-disclaimers - "As-Of" filtering - Demo dataset polish

15. Future Enhancements (Post-MVP)

- **Call Whisperer:** Real-time audio transcription during client phone calls to passively pull up relevant fund factsheets.
- **Automated CRM Sync:** Two-way integration with Salesforce and Outlook to fact-check drafted emails.
- **Multi-Document Comparative Matrix:** Side-by-side comparative table generation for up to 4 funds.
- **Cryptographic Audit Log:** Logging document citation hashes to a low-cost, tamper-proof ledger.

16. Success Metrics & KPIs

- **Faithfulness Rate:** $\geq 98\%$ of evaluated responses pass source verification.
- **Citation Accuracy:** 100% of generated claims have clickable citations pointing to valid coordinates.
- **Verification Speedup:** Average document look-up time reduced from 5 minutes to less than 15 seconds.
- **Advisor Usability Score:** System Usability Scale (SUS) score above 85.

17. Final Product Statement

VeriFund AI bridges the critical gap between complex financial documentation and compliant client advisory.

By unifying **tree-structured hierarchical retrieval**, **visual PDF grounding**, and **automated regulatory compliance checks**, VeriFund AI enables financial advisors to deliver rapid, high-confidence guidance while eliminating compliance risk and document lookup friction.